

SHAMIL K V

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Education

BS-MS in Mathematics (Minor in Economic Sciences) August 2019 – June 2025
Indian Institute of Science Education and Research (IISER), Bhopal **CGPA: 7.55**
Relevant Courses: Econometrics, Probability Theory and Markov Processes, Numerical Analysis, Microeconomics, Macroeconomics, Applied Production Analysis

Diploma in Data Science April 2025
Indian Institute of Technology (IIT) Madras **CGPA: 8.09**

Certifications

NISM Series VIII: Equity Derivatives Certificate Aug 2025
National Institute of Securities Markets (NISM) Score: 85.25%

Experience

Pashupati Capital Services Private Limited, Mumbai (Goregaon) Aug 2025 – Present
Quantitative Researcher

- Performed statistical and time-series analysis on **mid-frequency NSE equities, options, and futures data**, including **Nifty and Sensex derivatives**, to study volatility dynamics, liquidity, and regime behaviour.
- Developed and evaluated a focused set of **options strategies** using robust backtesting, out-of-sample validation, transaction cost modeling, and drawdown analysis, prioritizing stability over raw PnL.
- Implemented an automated **paper trading and execution-testing framework** using the **XTS API** to assess execution feasibility and calibrate slippage and fill assumptions.
- Designed and deployed **2 live strategies** driven by **volatility and open-interest signals**, and extended the research framework to additional signal classes.
- Architected a scalable **research and backtesting pipeline** using **InfluxDB, PostgreSQL, and pandas**, enabling multi-timeframe analysis and rapid hypothesis testing.

Technical Skills

Programming: Python, SQL, MATLAB, R, C, LaTeX

Databases: PostgreSQL, InfluxDB (time-series)

Tools & Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Excel, STATA

Finance: Black-Scholes, Binomial Trees, Duration, Convexity, Yield Curves, VaR, CVaR

Quant Techniques: Stochastic Differential Equations, Ito's Lemma, Monte Carlo Simulation, Time-Series Modeling, Backtesting of Systematic Strategies

Modeling: EDA, Supervised Learning, Reinforcement Learning, Feature Engineering

Quantitative Projects

Stochastic Optimal Control in Finance (Master's Thesis) 2024–2025
Supervisor: Dr. Somnath Pradhan

- Proved discrete-time Markov chain approximations for continuous-time portfolio optimization problems.
- Applied the **HJB equation** and dynamic programming to solve mean-variance and Merton optimal investment-consumption problems.
- Implemented **Q-learning** and policy iteration algorithms using NIFTY 50 and fixed deposit data for empirical validation.

Portfolio Optimization using Mean-Variance and Semi-Variance (L1-Penalized) 2025

- Implemented a **mean-variance portfolio optimization framework** with **downside risk control** using an **L1-penalized loss on negative returns**, preserving **convexity** and enabling **MINLP-based semi-variance models**.
- Built **multi-asset portfolios** comprising **NIFTY 500 equities, gold, silver, and index ETF funds** to improve **risk-adjusted returns** through diversification.
- Designed a **rolling-window optimization and rebalancing system**, analogous to a **feedback control mechanism**, with **weekly rebalancing** and **periodic re-optimization**, achieving approximately **7.6% annualized return**.

Data Analytics Projects

Bank Telemarketing Response Modeling 2024

- Developed and optimized classification models (**Logistic Regression, Random Forest, XGBoost**) to improve client targeting and lead conversion.
- Achieved 75% accuracy; evaluated performance using **ROC-AUC, Precision, Recall, and F1-score**.

Customer Segmentation, Retention & Pricing Optimization – B2B 2024

- Executed **quarterly and overall ABC segmentation** on revenue and normalized profit data; engineered weighted price reconstruction and derived **CLV and retention metrics** to identify persistent, seasonal, and high-impact customers.
- Conducted **Market Basket Analysis (Support, Confidence, Lift)** and developed a **linear demand-based profit maximization model** to design optimal bundles and data-driven pricing strategies.